



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Title: Title of the Project

BATCH ID : 23MPCS-D10

S. No	Roll. No	Student Name
1	23881A05FK	G. Spandana
2	23881A05GJ	B. Nithin Bharadwaj
3	23881A05GL	P. Bipin Raj

Guide Details

Sowmya M

Assistant professor
Department of CSE

Mini Project Outlines

- **Introduction**
- **Existing System Vs Proposed**
- **Research Gaps**
- **Problem Statement**
- **Proposed Project Objectives**
- **Proposed Project Outcomes**
- **Software and Hardware Requirements**
- **Architectural Diagram**
- **Proposed Methods and Algorithms**
- **Results**
- **Research Paper (Acceptance Proofs)**
- **Conclusion**

Introduction

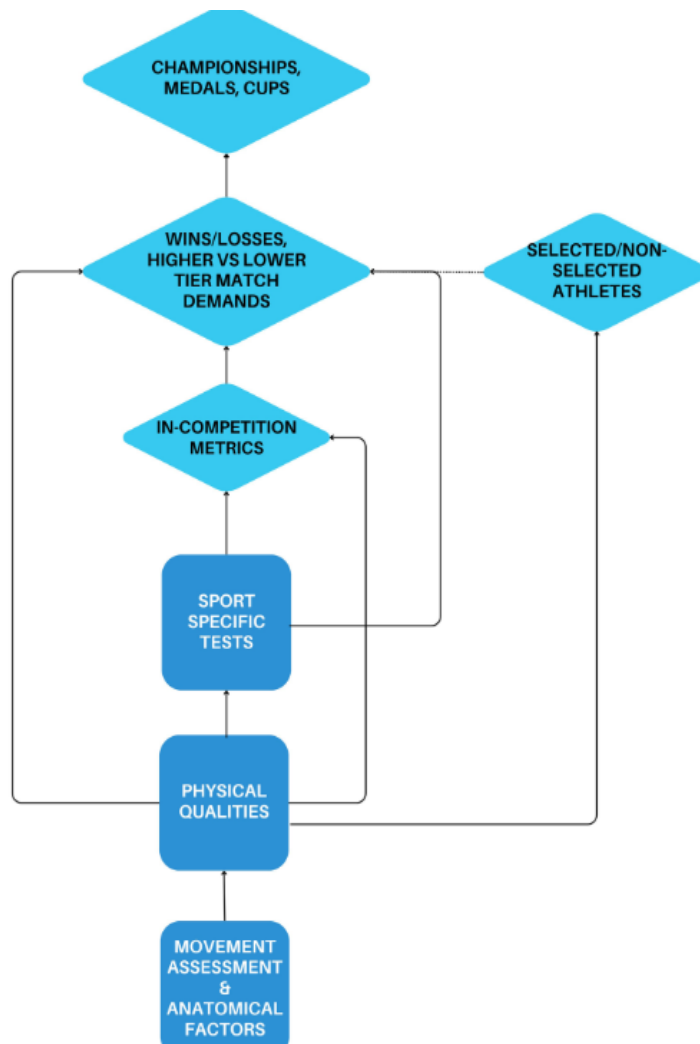
Traditional sports systems are expensive and not accessible to everyone. KhelQuest uses AI and smartphones to make talent identification fair, affordable, and available to all athletes.

- Sports talent identification is limited by **cost, location, and infrastructure**
- Traditional methods are **manual, biased, and inaccessible** to many athletes
- Advances in **AI and Computer Vision** enable automated performance analysis
- Smartphones can now be used for **low-cost sports evaluation**
- **KhelQuest** is an AI-powered platform that provides **fair, accessible, and real-time athlete assessment**



Existing System Vs Proposed

Existing System



Existing System Vs Proposed

Proposed System Architecture – KhelQuest *AI-Powered Platform for Democratizing Sports*

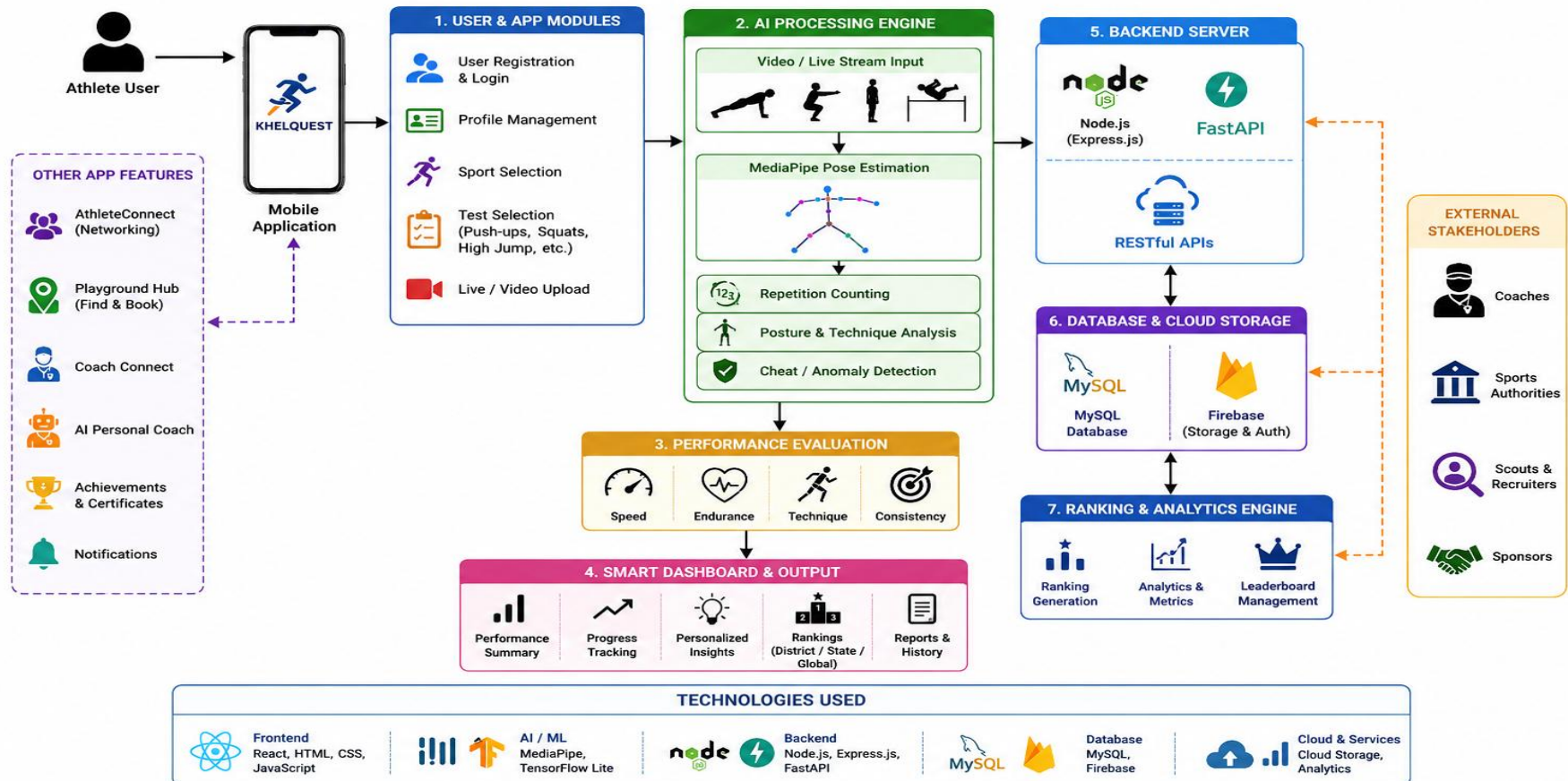
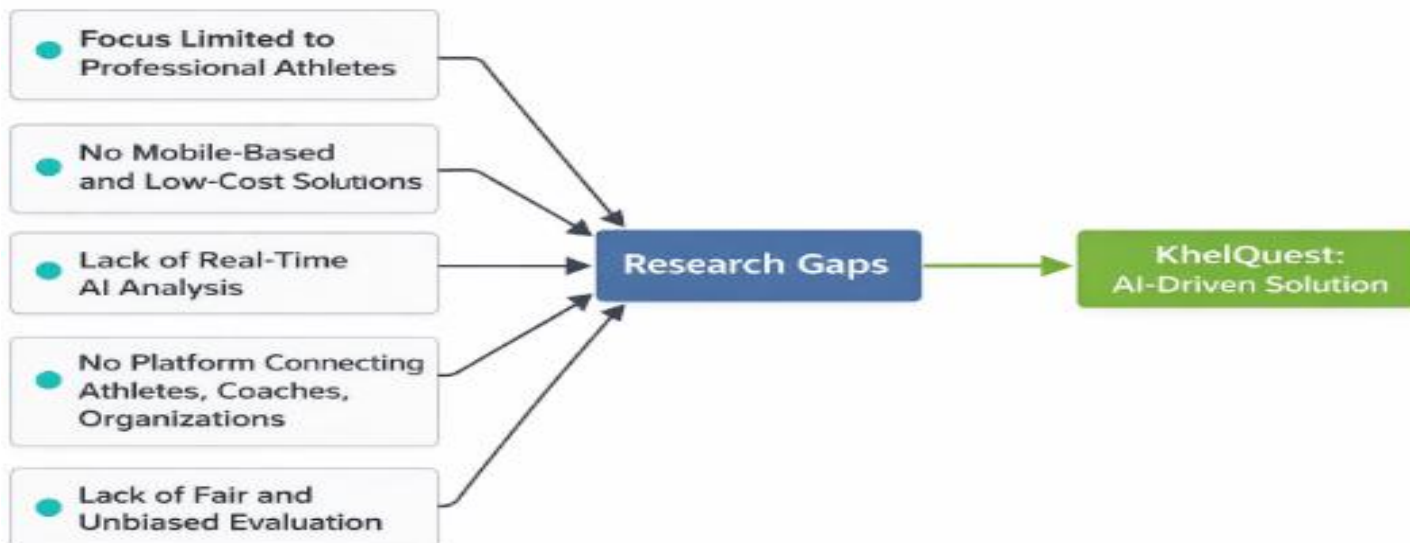


Fig. X. Proposed System Architecture of KhelQuest

Research Gaps

- Lack of mobile-based and low-cost solutions.
- Existing systems focus mainly on **professional athletes**, not grassroots level
- Limited use of **real-time AI-based performance analysis**
- No integrated platform connecting **athletes, coaches, and organizations**
- Existing methods rely on **expensive equipment and controlled environments**
- Insufficient focus on **fairness and unbiased evaluation**



Problem Statement

The major problem is that sports talent identification is not accessible or fair for everyone, especially for rural athletes, due to cost, bias, and lack of technology

- Lack of **equal access** to sports training and talent identification
- Traditional methods are **expensive and require specialized infrastructure**
- Evaluation is often **manual, subjective, and biased**
- Limited opportunities for **rural and underprivileged athletes**
- No scalable system for **real-time performance analysis and tracking**
- Absence of a unified platform connecting **athletes, coaches, and organizations**

Proposed Project Objectives

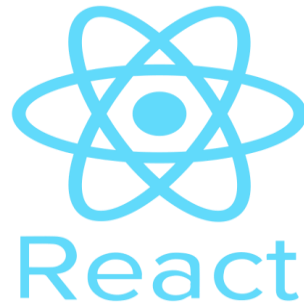
- To make sports talent identification accessible to all athletes using smartphones
- To replace expensive sports assessment equipment with AI-based video analysis
- To ensure fair and unbiased evaluation using AI-driven performance metrics
- To provide real-time feedback and performance improvement suggestions
- To develop personalized training plans based on athlete performance
- To enable athlete networking through the AthleteConnect feature
- To provide access to nearby playgrounds for practice and training
- To connect athletes with coaches (real and AI-based personal coach)
- To create a ranking system at district, state, and global levels
- To integrate sports authority access for trials, mentoring, and selection

Proposed Project Outcomes

- Increased access to sports talent identification for rural and underserved athletes
- Fair, transparent, and AI-based performance evaluation system
- Improved athlete performance through real-time feedback and training guidance
- Creation of a verified digital profile for every athlete
- Enhanced opportunities through rankings, coaching, and sports authority connections
- Strong athlete community through networking and collaboration
- Better decision-making for coaches and sports authorities using data-driven insights
- Reduced dependency on expensive sports assessment equipment

Software and Hardware Requirements

Software

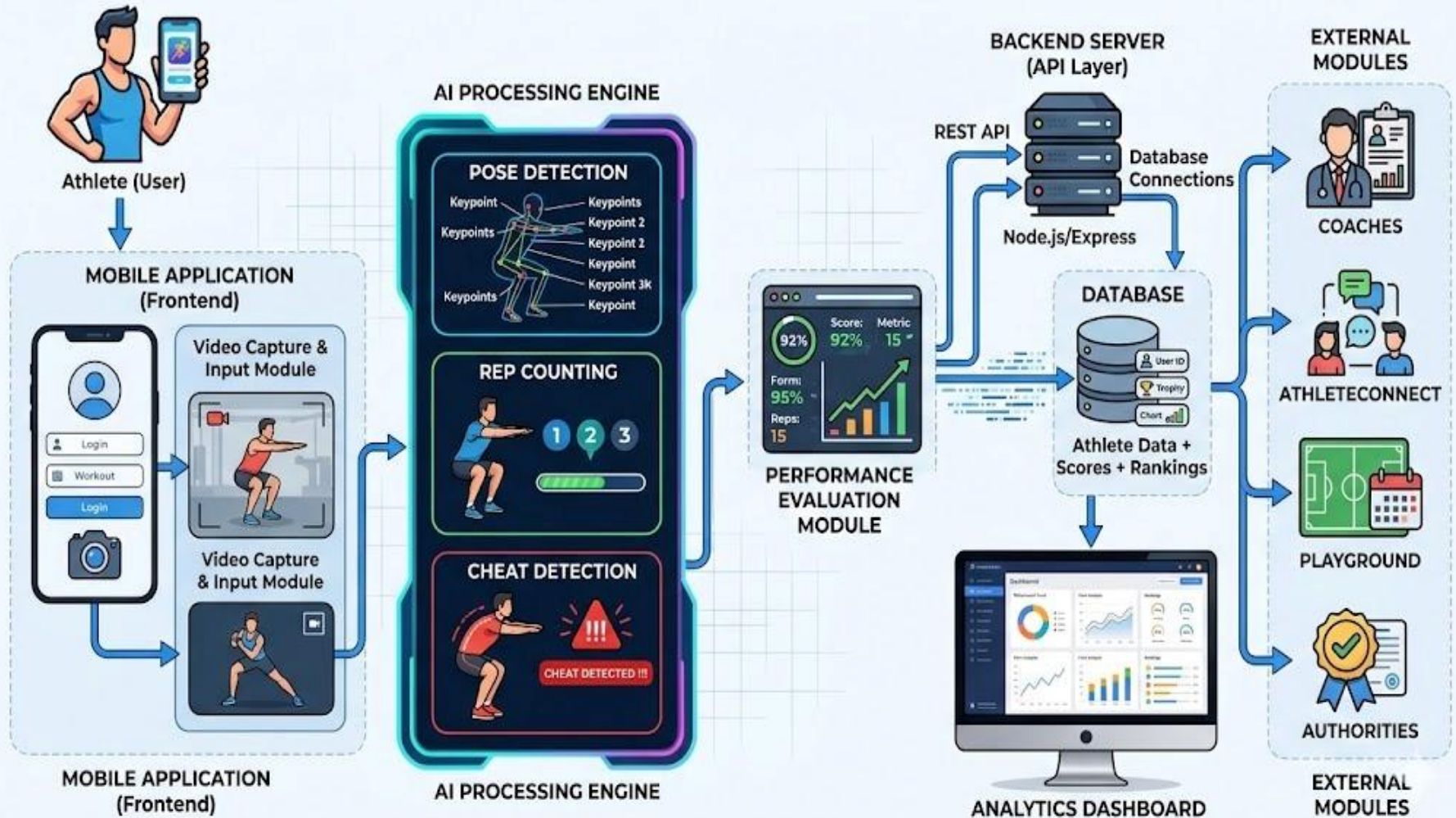


Hardware



Architectural Diagram

ATHLETE PERFORMANCE EVALUATION SYSTEM: ARCHITECTURE OVERVIEW



PROPOSED METHOD AND ALGORITHMS

Algorithms Used

1. Pose Estimation Algorithm

- MediaPipe / OpenCV
- Detects body joints (keypoints)
- Tracks movement in real-time

2. Rep Counting Algorithm

- Uses joint angles
- Detects motion cycles
- Counts repetitions accurately

3. Performance Evaluation Algorithm

Calculates:

- Speed
- Endurance
- Technique
- Consistency

4. Cheat Detection Algorithm

Identifies abnormal movement

Detects incomplete reps

5. AI Training Plan Generator

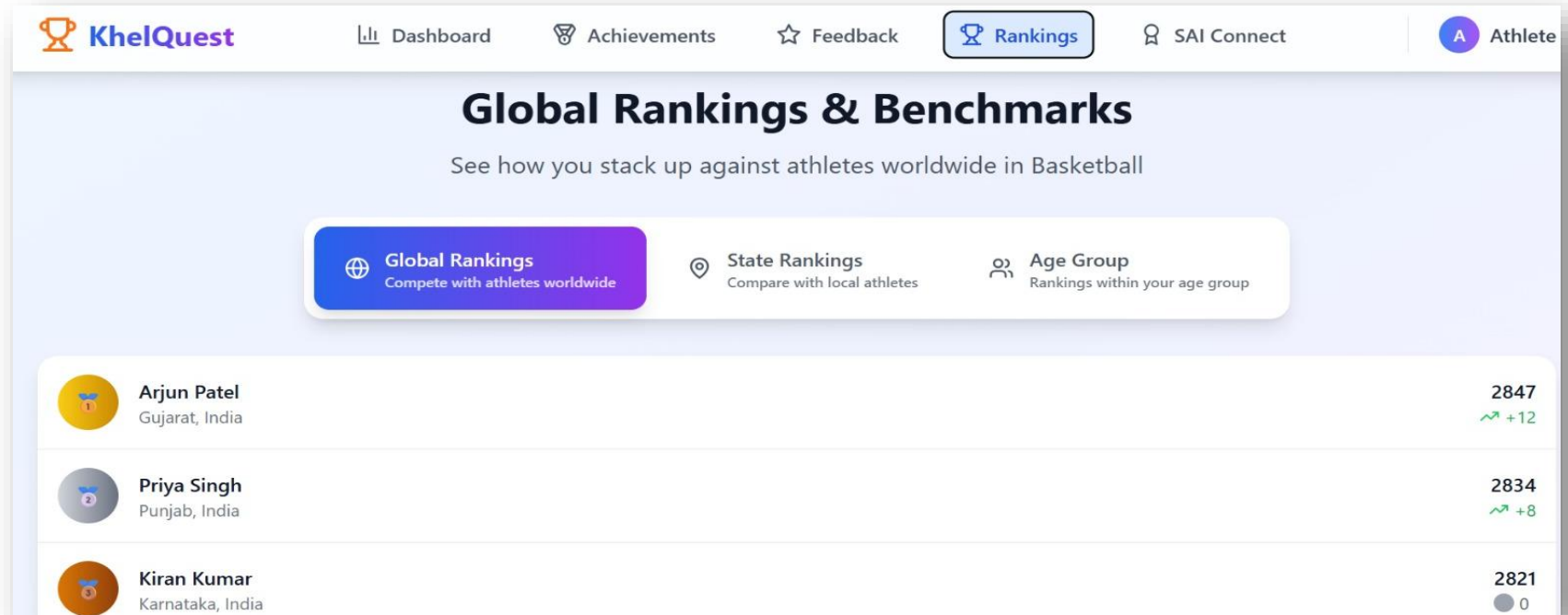
Rule-based + ML approach

Steps Involved:

- Capture video using mobile camera
- Extract frames from video
- Apply pose detection (MediaPipe)
- Identify body keypoints
- Analyze movement patterns
- Count repetitions (pushups, squats, jumps)
- Evaluate performance metrics
- Generate feedback and scores

Results

- Generated **real-time scores and feedback** for athletes
- Developed **ranking system** (state, age-group, global levels)
- Dashboard enables **performance tracking and comparison over time**
- System works using only **smartphone videos (no external sensors required)**



Research Paper (Acceptance Proof)

4/1/26, 9:17 PM

Conference Management Toolkit - Submission Summary

Submission Summary

Conference Name

1st International Conference on Next-Gen Computing: Converging Microelectronics and Mechatronics for Transformative Solutions

Paper ID

145

Paper Title

KhelQuest: AI-Powered Platform for Democratizing Sports

Abstract

The AI-Powered Platform for Democratizing Sports is designed to make sports talent identification accessible, affordable, and unbiased for athletes across all regions. Traditional methods of evaluating athletic performance often rely on expensive equipment, trained professionals, and centralized facilities, which limits opportunities for many talented individuals, especially in rural and underserved areas.

This project proposes a mobile-based solution that uses Artificial Intelligence and Computer Vision to analyze athlete performance through video input. By leveraging technologies such as pose estimation, the system can detect body movements, count repetitions for exercises like push-ups, squats, and jumps, and evaluate key performance metrics including speed, endurance, technique, and consistency.

The platform goes beyond assessment by providing real-time feedback, personalized training plans, and an AI-based personal coach to guide athletes in improving their performance. It also includes features such as athlete networking, playground discovery, coach connectivity, ranking systems at district/state/global levels, and integration with sports authorities for trials and mentoring opportunities.

Designed to work on low-end smartphones with minimal internet requirements, the system ensures inclusivity and scalability. By combining performance analysis, training support, and real-world opportunities, the proposed platform creates a complete ecosystem for athlete development and contributes to the democratization of sports talent identification.

Created

4/1/2026, 9:03:26 PM

Last Modified

4/1/2026, 9:03:26 PM

Authors

4/1/26, 9:17 PM

Conference Management Toolkit - Submission Summary

Spandana Gudikandula (Vardhaman college of engineering)

<spandanagudikandula1021@gmail.com>

Nithin Bharadwaj Batchu (Vardhaman college of engineering)

<nithinbharadwaj2005@gmail.com>

Bipin Raj Pathi (Vardhaman college of engineering) <bipinpathi@gmail.com>

Sowmya Mandala (Vardhaman college of engineering) <sowmyamandala@email.com>

Primary Subject Area

Next-Gen Intelligent Computing and Pervasive Networking

Submission Files

KhelQuest.pdf (786.9 Kb, 4/1/2026, 9:03:04 PM)

Research Paper (Acceptance Proof)

Submission Summary

Conference Name

2nd International conference on Computing, Communication and Green engineering

Track Name

Artificial Intelligence and Machine Learning

Paper ID

1371

Paper Title

KhelQuest: AI-Powered Platform for Democratizing Sports

Abstract

Access to structured sports opportunities remains uneven due to geographic, economic, and infrastructural constraints, particularly affecting emerging athletes and grassroots communities. This paper introduces KhelQuest, an AI-powered digital platform developed to promote inclusive and transparent sports participation by connecting athletes, coaches, organizers, and institutions within a unified ecosystem.

The platform utilizes artificial intelligence to support talent identification, personalized performance insights, event discovery, and systematic progress tracking. By reducing barriers related to cost, location, and limited professional networks, KhelQuest enables broader access to structured sports development.

Through data-driven evaluation and automated performance analysis, the system encourages fair talent assessment and informed decision-making for both athletes and stakeholders. Designed with scalability, accessibility, and transparency in mind, the proposed framework aims to transform how sports opportunities are identified and managed across decentralized environments.

This study presents the conceptual design, key functionalities, and potential societal impact of KhelQuest, emphasizing its role in fostering equitable participation and sustainable growth within the sports ecosystem.

Created

2/28/2026, 4:14:44 PM

Last Modified

2/28/2026, 4:25:42 PM

Authors

Spandana Gudikandula (Vardhaman college of engineering)

<spandanagudikandula1021@gmail.com>

Pathi Bipin Raj (Vardhaman college of engineering) <bipinpathi@gmail.com>

M Sowmya (Vardhaman college of engineering) <sowmyamandala@email.com>

Batchu Nithin Bharadwaj (Vardhaman college of engineering)

<bnithinbharadwaj2005@gmail.com>

Submission Files

KHELQUEST.pdf (785.2 Kb, 2/28/2026, 4:25:36 PM)

Research Paper (Acceptance Proof)

3/6/26, 12:18 PM

Conference Management Toolkit - Submission Summary

Submission Summary

Conference Name

2026 International Conference on Sustainable AI for Social Impact and Global Development

Track Name

Regular Paper

Paper ID

1005

Paper Title

KhelQuest: AI-Powered Platform for Democratizing Sports

Abstract

Access to structured sports opportunities remains uneven due to geographic, economic, and infrastructural constraints, particularly affecting emerging athletes and grassroots communities. This paper introduces KhelQuest, an AI-powered digital platform developed to promote inclusive and transparent sports participation by connecting athletes, coaches, organizers, and institutions within a unified ecosystem.

The platform utilizes artificial intelligence to support talent identification, personalized performance insights, event discovery, and systematic progress tracking. By reducing barriers related to cost, location, and limited professional networks, KhelQuest enables broader access to structured sports development.

Through data-driven evaluation and automated performance analysis, the system encourages fair talent assessment and informed decision-making for both athletes and stakeholders. Designed with scalability, accessibility, and transparency in mind, the proposed framework aims to transform how sports opportunities are identified and managed across decentralized environments.

This study presents the conceptual design, key functionalities, and potential societal impact of KhelQuest, emphasizing its role in fostering equitable participation and sustainable growth within the sports ecosystem.

Created

3/6/2026, 12:18:40 PM

Last Modified

3/6/2026, 12:18:40 PM

Authors

Spandana Gudikandula (Vardhaman college of engineering)

<spandanagudikandula1021@gmail.com>

Nithin Bharadwaj Batchu (Vardhaman college of engineering)

<bnithinbharadwaj2005@gmail.com>

3/6/26, 12:18 PM

Conference Management Toolkit - Submission Summary

Sowmya M (Vardhaman college of engineering) <sowmyamandala@email.com>

Bipin Raj Pathi (Vardhaman college of engineering) <bipinpathi@gmail.com>

Primary Subject Area

AI and Inclusive Development

Secondary Subject Areas

AI for Economic Empowerment

Submission Files

KHELQUEST.pdf (785,2 Kb, 3/6/2026, 12:18:19 PM)

Submission Questions Response

1. I confirm that this submission complies with the IEEE guidelines and plagiarism policies.

Please check the box below

Agreement accepted

2. I understand that if accepted, at least one author must register and present the paper.

Please check the box below

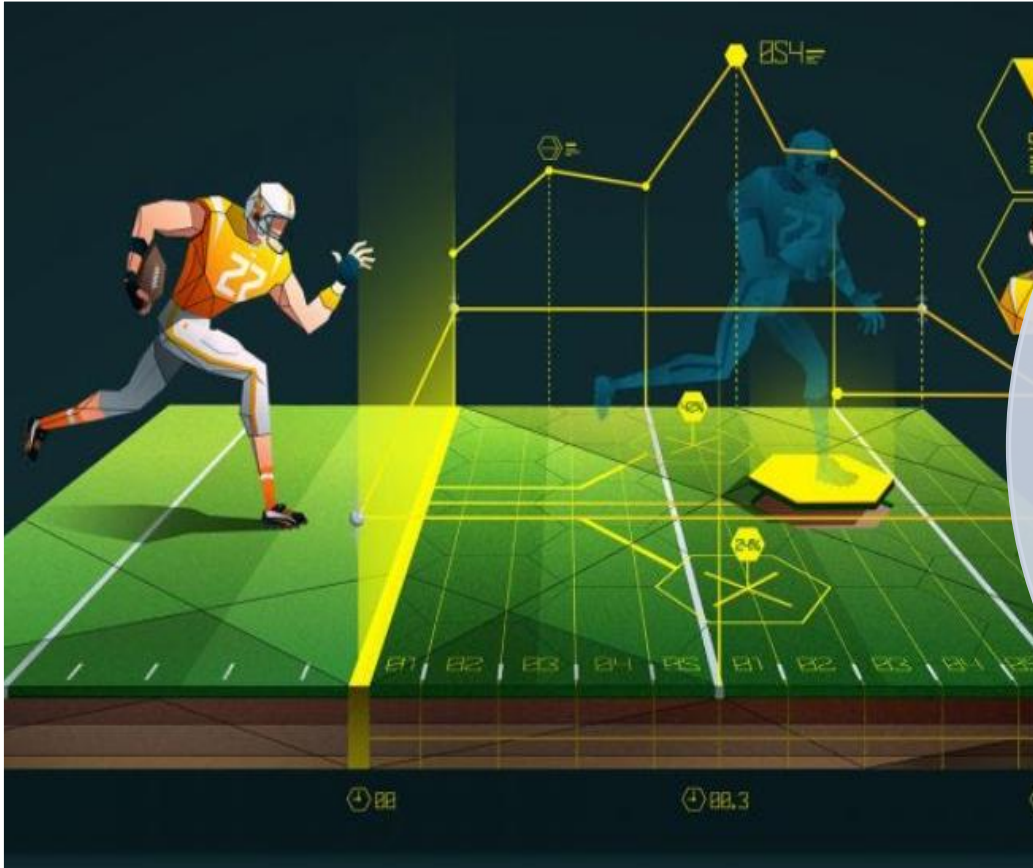
Agreement accepted

3. Disclosure on use of Generative AI

Authors understand that the use of artificial intelligence (AI)-generated content in an article (including but not limited to text, figures, images, and code) shall be disclosed in the acknowledgments section of any paper submitted to an IEEE Conference or Periodical. The sections of the paper that use AI-generated content shall have a citation to the AI system used to generate the content.

Agreement accepted

Conclusion



KhelQuest provides an AI-based solution for sports talent identification.

Reduces **human bias** through automated and standardized evaluation.

Supports **real-time feedback, ranking, performance tracking.**

Bridges the gap between **technology and grassroots sports development.**

Enables **low-cost and accessible performance analysis** using smartphones.

